Congenital Factor XII (Hageman Factor) Deficiency — Comprehensive Disease Characteristics Report

Disease: Congenital Factor XII Deficiency (Hageman trait) **MONDO ID:** MONDO:0009315 · **OMIM:** 234000 · **Orphanet:** ORPHA:330 · **ICD-10:** D68.2 · **ICD-11:** 3B10.1 · **MeSH:** D005173 (Factor XII Deficiency) **Category:** Mendelian (autosomal recessive) **Causal gene:** *F12* (HGNC:3530; locus 5q35.3; UniProt P00748)

Summary

Congenital Factor XII deficiency is the clearest example in human medicine of a dramatic laboratory abnormality that carries essentially no clinical consequence. It is an autosomal-recessive disorder caused by biallelic loss-of-function variants in the *F12* gene on chromosome 5q35.3. The hallmark is a **markedly prolonged activated partial thromboplastin time (aPTT)**—often exceeding 150–180 seconds against a reference range of roughly 22–32 s—accompanied by severely reduced FXII coagulant activity (FXII:C frequently <1–4%), yet **no bleeding tendency whatsoever**. This paradox is the central story of the disease: FXII initiates the contact (intrinsic) coagulation pathway *in vitro* but is dispensable for *in vivo* hemostasis. Most patients are identified incidentally during preoperative screening, during workup of an unexplained prolonged aPTT, or through family cascade testing, and require no treatment.

The mechanistic basis is now well understood. FXII is the initiator of the plasma **contact system**: it autoactivates on anionic surfaces (polyphosphate, extracellular DNA/NETs, collagen, and artificial surfaces such as kaolin or glass) to FXIIa, which drives two downstream arms—(1) the **intrinsic coagulation pathway** via activation of FXI, and (2) the **kallikrein-kinin system**, generating the vasoactive peptide bradykinin. Because the contact system contributes to thrombosis and inflammation (immunothrombosis) but not to the hemostatic plug that stops bleeding, its congenital absence is thrombo-neutral-to-protective and hemostatically silent. This decoupling has been confirmed across human cohorts, naturally FXII-deficient cats,

knockout and humanized-knock-in mouse models, and—most decisively—by the 2025 FDA approval of the anti-FXIIa antibody **garadacimab**, which pharmacologically recapitulates the deficiency safely in humans.

Two important counterpoints frame the entry. First, well-powered studies **refute** the older suspicions that FXII deficiency causes venous/arterial thrombosis or recurrent pregnancy loss; both associations fail to survive rigorous, adjusted analysis. Second, *F12* is a two-faced gene: while loss-of-function causes benign deficiency, **gain-of-function** variants (e.g., p.Thr328Lys) cause **hereditary angioedema with normal C1-inhibitor (HAE-FXII)**—a mechanistically opposite, estrogen-dependent, bradykinin-driven swelling disorder. The same enzyme that is harmless to lose is dangerous to over-activate.

Section 1 — Disease Information

Overview. Congenital FXII (Hageman factor) deficiency is a rare inherited coagulation-factor disorder characterized by reduced or absent plasma FXII activity. It is almost always **asymptomatic** and is typically discovered incidentally when a prolonged aPTT is found during routine or preoperative coagulation screening. Despite the striking *in vitro* clotting abnormality, affected individuals do not bleed abnormally, even during major surgery (PMID: 35417978; PMID: 41977511).

Key identifiers. MONDO:0009315 · OMIM 234000 · Orphanet ORPHA:330 · ICD-10 D68.2 ("Other specified coagulation defects") · ICD-11 3B10.1 · MeSH D005173.

Synonyms / alternative names. Hageman factor deficiency; Hageman trait; FXII deficiency; F12 deficiency; coagulation factor XII deficiency. (The disorder is named for John Hageman, the first patient described, whose prolonged clotting time was discovered incidentally without any bleeding history.)

Source of information. Characterization derives from **aggregated disease-level resources** (OMIM, Orphanet, systematic reviews) supplemented by **individual-patient and pedigree reports** (case reports, single-center cohorts with molecular genotyping). Both evidence streams converge on the same benign phenotype.

Section 2 — Etiology

Primary cause — genetic. The disease is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *F12***. Finding **F003** established this recessive basis: "Factor XII deficiency can be related to either homozygous or compound heterozygous pathogenic variants in the F12 gene" (PMID: 38057855). Heterozygous carriers exhibit intermediate FXII:C (~50%) and are generally considered to have partial deficiency rather than overt disease.

Genetic risk factors / modifier. The common **F12 46C>T promoter polymorphism (c.-4C>T, rs1801020)** is a quantitative modifier of plasma FXII levels. T-allele carriers have dose-dependently lower FXII: ~132 U/dL in 46CC, 87 U/dL in 46CT, and 53 U/dL in 46TT individuals (PMID: 18021303). This polymorphism explains much of the population variance in FXII activity and contributes to "partial" deficiency phenotypes.

Environmental / acquired factors. Congenital deficiency is genetically determined and not caused by environmental exposure. However, **acquired low FXII** can occur with liver disease (FXII is liver-synthesized), nephrotic syndrome, DIC, and in the presence of **anti-FXII autoantibodies or lupus anticoagulant / antiphospholipid antibodies**, which must be distinguished from inherited deficiency. Mild reductions in FXII:C have been noted in asymptomatic pediatric patients with transient antiphospholipid antibodies, so a differential diagnosis with mild congenital FXII deficiency should be considered (PMID: 11333473).

Protective factors / gene–environment interactions. Not applicable in the conventional sense. Paradoxically, the loss-of-function state is itself *protective* against thrombosis in model systems (see Sections 6 and 12), but there are no established dietary, lifestyle, or protective-allele factors that modify the benign congenital phenotype.

Section 3 — Phenotypes

The defining feature of FXII deficiency is the **dissociation between a laboratory abnormality and clinical silence**.

Phenotype	Type	Onset	Severity	Frequency	HPO suggestion
Prolonged aPTT	Laboratory abnormality	Congenital (detectable lifelong)	Marked (often >150 s)	~100% of severe cases	HP:0003645 (Prolonged partial thromboplastin time)
Reduced FXII:C / FXII:Ag	Laboratory abnormality	Congenital	Severe (<1–4% in homozygotes; ~50% in heterozygotes)	~100%	HP:0003005 (Abnormal coagulation factor activity)
Normal PT, bleeding time, platelets	Laboratory finding	Congenital	—	~100%	—
Absence of bleeding	Clinical (negative)	—	None	~100%	—

Characteristics. The abnormality is **congenital, stable, non-progressive, and lifelong**. In the Chinese Han cohort (n=14), all patients exhibited prolonged aPTT (mean 155.9 ± 89.7 s) and severely reduced FXII activity (mean 3.8 ± 4.0%) (PMID: 42636605, Finding **F001**). Crucially, this did not translate into bleeding: "Congenital FXII deficiency does not increase bleeding risk" (PMID: 41977511), and patients "do not experience a higher risk of surgical bleeding despite a markedly prolonged activated partial thromboplastin time" (PMID: 35417978).

Quality-of-life impact. For the great majority, essentially **none**. The disease does not impair daily functioning. The only meaningful QoL impacts are iatrogenic/psychological: unnecessary delay or cancellation of surgery, unnecessary transfusion of fresh frozen plasma, anticoagulation-management confusion (the prolonged aPTT cannot be used to monitor heparin), and anxiety from an abnormal lab value. No disease-specific EQ-5D/SF-36 data exist because there is no symptomatic disease to measure.

Section 4 — Genetic / Molecular Information

Causal gene. *F12* (HGNC:3530), on **chromosome 5q35.3**, encoding coagulation factor XII (Hageman factor), UniProt **P00748**. OMIM gene 610619; OMIM phenotype 234000.

Pathogenic variant spectrum. Severe deficiency arises from biallelic loss-of-function variants spanning missense, frameshift, and nonsense classes (Finding **F003**). Documented examples include:

Variant (protein)	Variant (cDNA)	Type	Zygosity / context	PMID
p.Gly506Asp	—	Missense	Homozygous	38057855
p.Lys365Glnfs*69	c.1092dup / c.1092_1093insC	Frameshift	Homozygous in 28.6% of a Chinese cohort; recurrent	42636605, 37730224
p.Cys238Leufs*73 + p.Glu521Lys	c.712_713insT + c.1561G>A	Frameshift + missense	Compound heterozygous	40372218
p.Phe51Serfs*44	c.150delC	Frameshift	Homozygous, consanguineous	37391120
p.Val579Hisfs*32	c.1792_1796delGTCTA	Frameshift	Compound heterozygous (pathogenic, PVS1)	37730224
p.Cys467Tyr	c.1457G>A	Missense (secretion defect)	Heterozygous modifier	41811038

Fifteen distinct *F12* variants (including eight novel) were catalogued in a single Chinese Han cohort, with impaired cellular secretion demonstrated for p.Q254P, p.S528Y, p.G376W, and p.V211Wfs*40: "Fifteen distinct F12 variants were identified, including eight novel ones. The frameshift variant c.1092dup was homozygous in 28.6% of this single-center cohort" (PMID: 42636605). In-vitro expression work shows that some missense variants (e.g., p.Cys467Tyr) reduce FXII plasma levels through a secretion defect*—protein accumulates in the cell lysate but fails to reach the supernatant—rather than through a transcription defect (PMID: 41811038).

Functional consequence. Loss of function (quantitative reduction in secreted, functional FXII). Origin is **germline**. There is no somatic/oncologic dimension to this disorder, so COSMIC/TCGA are not applicable.

Variant classification. Truncating variants are classified pathogenic under ACMG/AMP (PVS1-based); recurrent missense/secretion-defect variants are pathogenic/likely-pathogenic with functional support. The 46C>T promoter variant is a benign common polymorphism acting as a quantitative modifier.

Contrast — gain-of-function (HAE-FXII). The missense **p.Thr328Lys (c.1032C>A)** and related exon-9 variants are **gain-of-function** changes that increase FXII amidolytic activity without lowering plasma FXII levels, causing hereditary angioedema rather than deficiency (Finding **F004**; see Section 6).

Modifier genes / epigenetics / chromosomal abnormalities. The dominant modifier is the *F12* 46C>T promoter polymorphism itself. No disease-specific epigenetic signatures (methylation/histone) or chromosomal abnormalities (aneuploidy, translocation) are associated with congenital FXII deficiency; it is a single-gene, point-/small-indel disorder.

Section 5 — Environmental Information

Congenital FXII deficiency is a **purely genetic (Mendelian) disorder** with no environmental, lifestyle, or infectious etiology. There are no toxins, occupational exposures, dietary factors, or pathogens that cause the inherited condition. (Acquired FXII reduction—from liver disease, nephrotic syndrome, consumptive coagulopathy, or anti-FXII antibodies—is a separate, non-congenital phenomenon and a key differential diagnosis.) Infectious agents are not applicable.

Section 6 — Mechanism / Pathophysiology

Causal chain (ordered)

1. **Biallelic loss-of-function variant in *F12* → leads to** reduced synthesis/secretion of functional plasma FXII zymogen (demonstrated: truncating variants, secretion-defect missense variants; PMID: 42636605, PMID: 41811038).
2. Low/absent plasma FXII → **results in** failure of contact activation when blood contacts an anionic surface (kaolin, glass, ellagic acid) in the aPTT reagent → **leads to** a **markedly prolonged aPTT** with normal PT (demonstrated; PMID: 30630206).

3. Branch A (in vitro / artificial surfaces): Loss of FXIIa → **results in** no downstream FXI activation in the test tube → prolonged clotting time. **BUT** — the branch does **not** propagate to the body: in vivo hemostasis proceeds normally via tissue factor/FVIIa (extrinsic) and FXI activation by thrombin, so **no bleeding results** (demonstrated across cohorts and surgery; PMID: 41977511, PMID: 35417978).
4. Branch B (pathological surfaces in vivo—polyphosphate, NETs/DNA): In normal individuals FXII autoactivates here to drive **immunothrombosis** and **bradykinin** generation; in deficiency this branch is **blunted**, which is thrombo-protective and anti-inflammatory rather than harmful (demonstrated in models; PMID: 29483100, PMID: 34363738, PMID: 41683756).
5. Net clinical manifestation → a **benign, asymptomatic prolonged aPTT**, usually discovered incidentally.

Normal FXII biology and the decoupling principle

FXII is a **modular multidomain zymogen** (Finding **F007**). Its heavy chain comprises fibronectin type II and type I domains, two EGF-like domains, a kringle domain, and a proline-rich region; the light chain is the serine-protease catalytic domain. On contacting a negatively charged surface, FXII binds and **autoactivates** to FXIIa. Surface binding maps to the **fibronectin type II domain** (residues ~30–47; PMID: 11154114) and, critically, to **two positively charged lysine/histidine patches in the EGF-1 domain**: "Two unique, positively charged patches in the EGF-1 domain were identified (upstream, 73K74K76K78H81K82H; downstream, 87K113K). Neutralizing the charge of both patches led to a 99% reduction in FXII kaolin binding, subsequent decrease in autoactivation of 94%" (PMID: 38897387)—with aPTT prolonged from 36 to 223 s, a structural proof that surface binding is the trigger step.

FXIIa then drives **two pathways** (Finding **F005**): "This system activates 2 distinct pathways, the intrinsic pathway of coagulation via cleavage of FIX, and inflammation via HK cleavage resulting in bradykinin (BK) generation" (PMID: 40638675). Beyond proteolysis, zymogen FXII "signals via urokinase-type plasminogen activator receptor (uPAR) to induce neutrophil extracellular trap formation (NETosis), thereby amplifying immunothrombosis" (PMID: 41301820).

The pivotal mechanistic insight—explaining the entire benign phenotype—is that **thrombosis and hemostasis are decoupled at FXII**: "targeting factor XII or its activator polyphosphate, provides protection from thromboembolic diseases without interfering with hemostasis" (PMID: 29483100), and in animal models "either its knock-out or inhibition in animal models efficiently reduced thrombosis without affecting normal hemostasis" (PMID: 34363738).

Gain-of-function branch (the mechanistic opposite)

The same gene produces an opposite disease when activity is *increased*. The HAE-causing **p.Thr328Lys** is "a gain-of-function mutation that markedly increases FXII amidolytic activity but that does not alter FXII plasma levels" (PMID: 17186468). Excess FXIIa over-drives the kallikrein-kinin arm → excess **bradykinin** → recurrent angioedema—especially under high-estrogen states (oral contraceptives, pregnancy), almost exclusively in women; in some such families "mutations in the coagulation factor XII (Hageman factor) gene were detected" (PMID: 24176211) (Finding **F004**).

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                        ┌─ LOSS OF FUNCTION (deficiency) — benign, no bleeding, thrombo-neutral
F12 variant ────────────┤
                        └─ GAIN OF FUNCTION (p.Thr328Lys) — HAE-FXII, excess bradykinin, angioe

Normal FXII pathway:
anionic surface → FXII → FXIIa ──► (A) FXI → intrinsic coagulation  [in vitro aPTT; dispensab
                                ├─► (B) prekallikrein → kallikrein → HK → BRADYKININ  [inflar
                                └─► uPAR signaling → NETosis → immunothrombosis
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Ontology suggestions. GO:0007596 (blood coagulation); GO:0002542 (Factor XII activation); GO:0007597 (blood coagulation, intrinsic pathway); GO:0030193 (regulation of blood coagulation); CHEBI:bradykinin; CHEBI:polyphosphate. Cell types: hepatocyte (CL:0000182, source of FXII); neutrophil (CL:0000775, NETosis/immunothrombosis); platelet (CL:0000233, polyphosphate source).

Section 7 — Anatomical Structures Affected

Congenital FXII deficiency is a **systemic plasma-protein deficiency without a target organ**; there is no structural tissue pathology.

- **Organ / system level.** The affected "compartment" is the **blood / circulatory system** (UBERON:0000178, blood; UBERON:0001969, blood plasma). The **liver** (UBERON:0002107) is the site of FXII synthesis and thus the biological origin of the deficient protein, but the liver is structurally normal. No organ is damaged by the deficiency.
- **Tissue / cell level. Hepatocytes** (CL:0000182) synthesize FXII. Cells functionally relevant to normal FXII biology (and therefore to its absence) include **neutrophils** (CL:0000775,

NETosis/immunothrombosis) and **platelets** (CL:0000233, polyphosphate release). None is pathologically altered in deficiency.

- **Subcellular level.** FXII is a **secreted plasma protein**; relevant compartments are the **endoplasmic reticulum / secretory pathway** (GO:0005783; GO:0005576, extracellular region) where secretion-defect missense variants cause intracellular retention (PMID: 41811038).
- **Localization / lateralization.** Not applicable—this is a diffuse plasma deficiency with no anatomical site or lateralization.

Section 8 — Temporal Development

- **Onset. Congenital**; the deficiency is present from birth and is lifelong. Because it is asymptomatic, the *age of detection* varies widely—from neonatal incidental findings to late-adult preoperative workups.
- **Onset pattern.** Chronic/constitutive (a stable trait, not an acute event).
- **Progression. None.** The disorder is **stable and non-progressive**; FXII levels remain constant throughout life (barring superimposed acquired causes). There are no disease "stages."
- **Disease course.** Chronic and lifelong, but clinically silent. There is no relapsing-remitting or progressive course.
- **Remission / critical periods.** Not applicable. The only clinically relevant "time windows" are iatrogenic decision points—e.g., **perioperative assessment**, where the prolonged aPTT must be correctly interpreted as benign to avoid unnecessary intervention, and **pregnancy/delivery**, where management guidance is debated but the deficiency itself does not warrant routine prophylaxis.

Section 9 — Inheritance and Population

Inheritance. Autosomal recessive. Severe (complete) deficiency requires biallelic variants; heterozygotes have partial (~50%) deficiency that is itself benign (Finding **F003**). Consanguinity increases the likelihood of homozygous severe deficiency—documented in a consanguineous family where "Sequencing revealed a homozygous frameshift mutation c.150delC (p.Phe51Serfs*44) site in the F12 gene exon 3" (PMID: 37391120).

Penetrance / expressivity. The laboratory phenotype (prolonged aPTT, low FXII:C) is **fully penetrant** for the biochemical trait; the *clinical* phenotype is essentially null (no symptoms). Expressivity of FXII level varies with genotype and the 46C>T modifier.

Epidemiology.

Measure	Estimate	Source
Severe (homozygous/complete) deficiency incidence	**≈ 1 in 1,000,000**	PMID: 35417978
Partial (heterozygous) deficiency	Relatively common	PMID: 35417978
Low FXII in a general population (Korea)	High incidence of plasma FXII deficiency reported in normal Korean subjects	PMID: 16165195

"A total absence of FXII as seen in homozygous patients, is rare, with an incidence of approximately 1/1,000,000 individuals" (PMID: 35417978, Finding **F006**).

Population demographics. Partial FXII deficiency / low FXII activity appears **more frequent in some East Asian populations** (notably a high incidence reported in normal Korean subjects, PMID: 16165195), consistent with the recurrent frameshift variants seen in Chinese Han cohorts (PMID: 42636605). The 46C>T promoter T-allele—lowering FXII—has substantial population frequency (~38% heterozygous, ~5% homozygous in one European cardiology cohort; PMID: 18021303). **Sex ratio** for the deficiency is approximately equal (autosomal). (By contrast, the gain-of-function HAE-FXII disorder occurs almost exclusively in women.)

Section 10 — Diagnostics

Core laboratory pattern (Finding F006):

1. **Isolated prolonged aPTT with normal PT** (intrinsic-pathway screen abnormal; extrinsic normal). "The APTT is also sensitive to contact factor deficiencies, including factor XII, prekallikrein, and high-molecular-weight kininogen" (PMID: 30630206).
2. **Mixing study corrects the aPTT**—indicating factor deficiency rather than an inhibitor: "an isolated prolonged aPTT, which was corrected with mixing studies, and a severe FXII deficiency with activity levels < 1%" (PMID: 41977339).
3. **Confirmation by one-stage FXII:C clotting assay** plus **FXII:Ag ELISA** to distinguish cross-reactive-material (CRM) status.
4. **Viscoelastic testing** shows a hypocoagulable INTEM with a normal EXTEM (PMID: 35417978).

Key differential diagnoses. Other contact-factor deficiencies (prekallikrein, high-molecular-weight kininogen), which also prolong the aPTT without bleeding (PMID: 30630206); **lupus anticoagulant / antiphospholipid antibodies** (which do *not* fully correct on mixing and may coexist with mildly reduced FXII:C, PMID: 11333473); heparin contamination; and bleeding-associated intrinsic-factor deficiencies (FVIII, FIX, FXI—which *do* cause bleeding and must be excluded).

Genetic / molecular confirmation. Sanger sequencing or whole-exome sequencing of *F12* confirms the diagnosis and identifies the causal variant(s) (PMID: 37391120; PMID: 42636605). Single-gene *F12* testing is sufficient; WGS/CMA/karyotyping/FISH/mtDNA/repeat-expansion testing are **not applicable**. A key clinical caveat: the prolonged aPTT **cannot be used to monitor heparin** in these patients and can confound activated-clotting-time (ACT) assessment during cardiac surgery (PMID: 35417978).

Omics. Plasma **metabolomic profiling** has been applied to FXII-deficient cohorts (PMID: 42636605), but no omics test is required for diagnosis.

Screening. There is **no newborn or population screening** for FXII deficiency (no clinical benefit). **Cascade family testing** is reasonable for genetic counseling once a proband is identified, chiefly to reassure relatives and prevent unnecessary perioperative intervention.

Section 11 — Outcome / Prognosis

Prognosis is excellent—essentially a normal life expectancy. FXII deficiency does not cause bleeding, does not shorten survival, and does not independently cause thrombosis.

Thrombosis—refuted as a causal risk (Finding F002). Despite longstanding speculation (fueled by the original patient John Hageman, who later died of a pulmonary embolism), rigorous studies find **no independent association**:

Study	Design	Result
Leiden Thrombophilia Study (PMID: 7947293)	Population-based case-control (DVT)	FXII deficiency 6% (21/350) cases vs 5% (18/350) controls; **matched OR 1.2 (95% CI 0.6–2.4)**. "We do not consider factor XII to be a determinant of deep-vein thrombosis."
2026 systematic review/ meta-analysis (PMID: 42035525)	6 studies, 5,003 individuals	Pooled **RR 1.34 (95% CI 0.99–1.81)**; DVT subgroup RR 1.44 (0.88–2.36); **adjusted OR 1.03 (0.65–1.64)** — not significant.

Lower FXII activity is best regarded as a **risk marker rather than a risk factor** for cardiovascular disease (PMID: 18331454); emerging consensus even views FXII (and FXI) deficiency as **protective** against thrombosis (PMID: 42610269; PMID: 41977511).

Recurrent pregnancy loss—controversial, not clearly causal (Finding F009). Some cohorts report reduced FXII in women with recurrent spontaneous abortion (PMID: 24597288) or recurrent IVF-ET failure (median FXII 79.5 vs 109, $p<0.001$; PMID: 18294355). However, a careful study found "Low FXII activity was not associated with RPL"—only the 46C/T genotype showed a weak association (OR 2.8), and FXII level did not predict subsequent miscarriage (PMID: 25489738). Where a reproductive association exists, it may reflect **acquired anti-FXII autoantibodies** (which augment platelet aggregation), not congenital deficiency (PMID: 26011374).

Complications / morbidity. The chief "complications" are **iatrogenic**: unnecessary surgical delay, inappropriate FFP transfusion, and misinterpretation of heparin monitoring. True disease-related morbidity is negligible.

Section 12 — Treatment

No treatment is required for congenital FXII deficiency itself. Because there is no bleeding tendency, factor replacement and prophylaxis are generally unnecessary even for major surgery (PMID: 35417978). Management is largely **educational**: correctly interpreting the prolonged aPTT, avoiding unneeded blood products, and counseling.

Perioperative / obstetric management is individualized and not standardized. A pregnancy case series (19 pregnancies across 9 publications) used prophylactic fresh frozen plasma with selective low-molecular-weight heparin, emphasizing multidisciplinary, dynamic risk stratification balancing a theoretical bleeding/thrombosis concern—though the evidence base is thin and guidelines are lacking (PMID: 41429486).

FXII as a therapeutic target (the inverse application). The deficiency's safety makes **FXIIa inhibition** an attractive antithrombotic/anti-inflammatory strategy (Findings **F005**, **F011**). Classes in development include antisense oligonucleotides, siRNAs, monoclonal antibodies, peptides (e.g., Ir-CPI), and recombinant inhibitors (rHA-Infestin-4) (PMID: 41640375; PMID: 41977511).

Clinical proof-of-concept — garadacimab (Finding F011). Garadacimab (Andembry, garadacimab-gxii) — "a first-in-class monoclonal antibody that inhibits activated Factor XII (FXIIa)" — was FDA-approved for "prophylaxis of HAE attacks in patients aged ≥12 years" (PMID: 42433832; PMID: 41499201).

Parameter	Value	Source
Pivotal trial	Phase 3 VANGUARD (NCT04656418)	PMID: 42433832
Attack reduction	**91.5%**; ~77% attack-free	PMID: 42433832
Integrated efficacy	94.9% monthly-attack reduction (n=164)	PMID: 42268494
Dosing	200 mg SC monthly (half-life ~18–20 d)	PMID: 42057602
Safety	No drug-related SAEs; mild injection-site reactions, headache, nasopharyngitis; no deaths	PMID: 42268494

Critically, the **long-term safety profile of pharmacologic FXIIa blockade mirrors the benign congenital deficiency**: preclinically, "FXII inhibition prevented thrombus formation and reduced fibrin deposition, achieving effects comparable to heparin but with markedly lower bleeding risk" (PMID: 41977511). Garadacimab thus provides a human "experiment" confirming that removing FXII(a) activity is safe. **NCIT suggestion:** Garadacimab (monoclonal antibody); drug class — anti-FXIIa monoclonal antibody / contact-pathway inhibitor.

Section 13 — Prevention

Because congenital FXII deficiency is a benign Mendelian trait, **classical prevention is neither needed nor possible**. There is no primary prevention (no modifiable risk factor), no immunization, and no indicated population screening.

- **Genetic counseling** is the only relevant preventive measure: explaining autosomal-recessive inheritance, offering **cascade testing** of relatives primarily to prevent *iatrogenic harm* (unnecessary surgical delay/transfusion), and providing reassurance regarding the absence of bleeding risk.
- **Tertiary "prevention"** amounts to avoiding inappropriate medical intervention based on the isolated prolonged aPTT.
- For affected women contemplating pregnancy, individualized counseling is reasonable given the (unresolved) literature on reproductive outcomes, but no standardized prophylaxis is established (PMID: 41429486).

Section 14 — Other Species / Natural Disease

FXII deficiency occurs **naturally and commonly in the domestic cat** (*Felis catus*; NCBI Taxon 9685), providing a striking comparative-biology confirmation of the benign phenotype (Finding **F010**).

- A clinically normal cat showed "an intrinsic coagulation defect characterized by prolonged activated partial thromboplastin time, with normal bleeding time, platelet count, and prothrombin time," confirmed as FXII deficiency (PMID: 879587).

- Breeding studies established an inheritance pattern that "revealed an autosomal recessive pattern similar to that reported in humans (Hageman trait)" (heterozygotes ~50% FXII:C; homozygotes <2%) (PMID: 7427778).
- A modern cohort found that "FXII deficiency is common in domestic cats, and also present in many different breeds," with a Midwestern-US geographic bias; **none of 26 deficient cats bled**, and two high-frequency *F12* variants were identified: "an exon 13 missense mutation (c.1631G > C) and an exon 11 deletion mutation (c.1321delC)" (PMID: 31022435).

Evolutionary perspective (Finding F008). An evolutionary review frames the FXII contact system as a **foreign-surface-sensing / immunothrombosis module** that arose in terrestrial animals, consistent with the minimal bleeding phenotype of deficiency and with the **secondary loss of *F12* in marine mammals**: these facts help "explain the minimal bleeding phenotype of FXII deficiency and the secondary loss of F12 in marine mammals" (PMID: 41301820). The gene is evolutionarily dispensable for hemostasis—whole lineages have lost it. **Orthologous gene:** cat *F12*; mouse *F12*. No zoonotic/transmission dimension applies.

Section 15 — Model Organisms

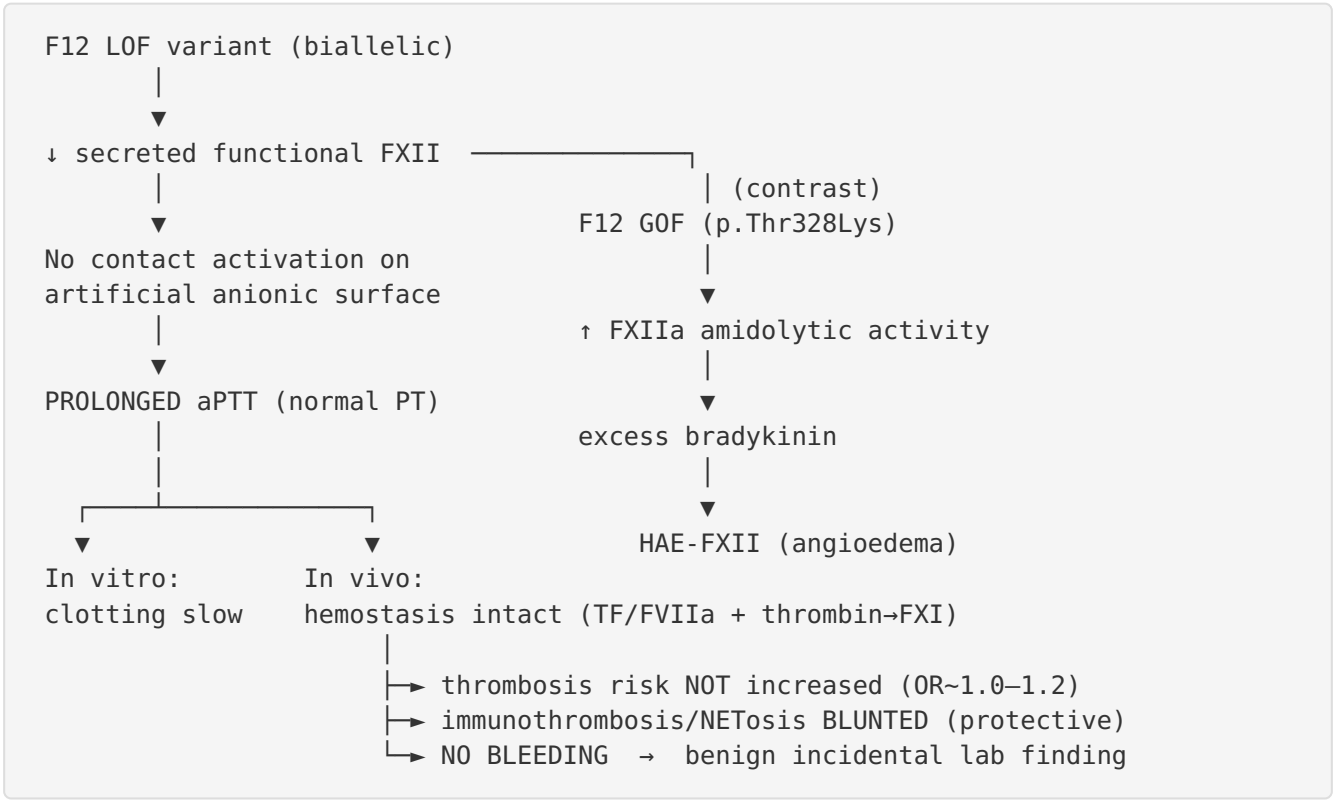
- **Mouse (*Mus musculus*, NCBI Taxon 10090).** *F12* **knockout mice** are the foundational model: they show reduced experimental thrombosis **without impaired hemostasis**, directly demonstrating the thrombosis/hemostasis decoupling (PMID: 34363738; PMID: 29483100). A **humanized FXII knock-in mouse** was generated as "a powerful model system to test novel anti-thrombotic agents," enabling testing of human-FXII-targeting therapeutics in vivo (PMID: 34363738).
- **Disease-relevant model phenotypes (Finding F012).** FXII deficiency/inhibition in rodent models reduces arterial/venous thrombosis and extracorporeal-circuit clotting; reduces **stroke infarct volume**; attenuates **neutrophil-driven inflammation**—"FXII deficiency has been associated with reduced neutrophil adhesion and migration in sterile skin wounds and immune complex-induced vasculitis"; and in neurological conditions "FXII deficiency significantly reduced the number and severity of multiple sclerosis relapses and decreased the volume of post-traumatic brain oedema" (PMID: 41683756).
- **Cat (natural model).** The naturally FXII-deficient domestic cat is an informative *spontaneous* model recapitulating the human recessive, non-bleeding trait (PMID: 31022435).

- **In vitro.** HEK293T expression systems have characterized the molecular consequences of *F12* missense variants (secretion defects), a tractable system for variant interpretation (PMID: 41811038).

Phenotype recapitulation. Mouse knockouts faithfully reproduce the human **non-bleeding** phenotype and the **anti-thrombotic/anti-inflammatory** consequences—arguably better than they model any "disease," since FXII deficiency has no overt human pathology to reproduce. **Limitation:** because the human condition is essentially asymptomatic, models are used to study FXII's *normal* roles and to validate it as a *drug target*, rather than to model a damaging disease.

Mechanistic Model / Interpretation (Synthesis)

The unifying thesis is the **thrombosis–hemostasis decoupling at FXII**, which explains every observation in this report:



Four independent evidence streams converge on the same conclusion: 1. **Human cohorts** — severe deficiency, markedly prolonged aPTT, no bleeding (PMID: 42636605, PMID: 35417978). 2. **Epidemiology** — no causal thrombosis link (PMID: 7947293, PMID: 42035525). 3. **Comparative biology** — cats deficient and non-bleeding; marine mammals have *lost* the gene (PMID: 31022435, PMID: 41301820). 4. **Pharmacology** — FXIIa blockade (garadacimab) is safe and effective in humans (PMID: 42433832).

The deeper insight is that FXII is better understood as an **immunothrombosis / foreign-surface-sensing node** than as a hemostatic factor. Its congenital absence removes a pathological-amplification pathway (thrombosis, NETosis, bradykinin-driven inflammation) while leaving physiological clotting intact—making "the disease" a near-perfect natural knockout and the gene an ideal drug target.

Evidence Base

PMID	Role in this report
42636605	Chinese Han cohort: aPTT 155.9±89.7 s, FXII:C 3.8±4.0%; 15 variants incl. recurrent c.1092dup; metabolomics (F001, F003, F006)
30630206	aPTT sensitivity to contact-factor deficiencies incl. FXII (F001)
41977511	FXII deficiency does not increase bleeding; FXII inhibition antithrombotic with low bleeding (F001, F002, F005, F011)
7947293	Leiden Thrombophilia Study — no DVT association (F002)
42035525	Meta-analysis — no significant thrombosis association (F002)
38057855	Recessive genetic basis; homozygous p.Gly506Asp (F003)
37391120	Consanguineous family, homozygous frameshift; WES (F003, F006)
40372218, 37730224	Compound-heterozygous pedigrees, variant spectrum (F003)
41811038	p.Cys467Tyr secretion-defect mechanism (Sections 4, 7)
17186468	p.Thr328Lys gain-of-function → HAE (F004)
24176211	F12 mutations in HAE with normal C1-INH (F004)
27905115	HAE-FXII treatment cohort (n=72) (F004)
40638675	Contact system structure; dual intrinsic + bradykinin pathways (F005)
29483100	Thrombosis/hemostasis decoupling (F005)
34363738	Humanized FXII knock-in mouse; KO reduces thrombosis, spares hemostasis (F005, Sec 15)
38897387	EGF-1 domain patches mediate surface binding/autoactivation (F007)
11154114	Fibronectin type II surface-binding site (F007)
35417978	Incidence ~1/1,000,000; no surgical bleeding; ACT/viscoelastic findings (F006, Sec 11)
41977339	Diagnostic pattern: isolated prolonged aPTT correcting on mixing, FXII <1% (F006)
41301820	Evolutionary immunothrombosis module; marine-mammal F12 loss; uPAR/NETosis (F008, F012)

PMID	Role in this report
25489738	RPL not associated with low FXII (F009)
24597288, 18294355, 26011374	Opposing RPL/IVF evidence; anti-FXII autoantibodies (F009)
31022435, 7427778, 879587	Feline natural disease: common, AR, non-bleeding; F12 variants (F010)
42433832, 42268494, 41499201, 42057602	Garadacimab FDA approval, efficacy, safety, dosing (F011)
41683756	FXII deficiency anti-inflammatory/neuroprotective in models (F012)
18021303, 18331454	46C>T polymorphism dose-effect; FXII as risk marker not factor (Sec 4, 11)
16165195	High incidence of low FXII in Korean subjects (Sec 9)
42610269	FXII/FXI deficiency viewed as thrombo-protective (Sec 11)
41429486	Pregnancy management case series (Sec 12, 13)
11333473	Mild FXII reduction vs antiphospholipid antibodies — differential dx (Sec 2, 10)

Evidence-source mix: human clinical cohorts/case reports and epidemiology (most sections); model-organism (mouse KO/knock-in, natural feline disease); in vitro (HEK293T variant expression, domain mutagenesis); and computational/evolutionary (comparative genomics of F12 loss).

Limitations and Knowledge Gaps

1. **Rarity limits power.** Complete (homozygous) deficiency is ~1/1,000,000, so cohorts are small (often single-center, n≤30) and geographically skewed (many from East Asian centers), limiting generalizability of variant-frequency and metabolomic data.
2. **Reproductive outcomes unresolved.** The link to recurrent pregnancy loss remains genuinely controversial. The best-designed study is negative, but publication bias and the

confound of acquired anti-FXII autoantibodies vs congenital deficiency are not fully disentangled.

3. **No standardized perioperative/obstetric guidelines.** Management in pregnancy and surgery is based on small case series and expert opinion, not trials.
4. **Thrombosis meta-analysis borderline.** Although the adjusted OR is ~1.0, the unadjusted pooled RR (1.34, CI 0.99–1.81) sits at the edge of significance, reflecting residual confounding and heterogeneity across studies.
5. **Genotype–phenotype quantitation.** Precise mapping of specific *F12* variants to FXII:C/FXII:Ag levels (CRM+ vs CRM−) and the modifying effect of 46C>T is incompletely catalogued.

Proposed Follow-up Experiments / Actions

1. **Harmonized international *F12* variant registry** linking genotype (including 46C>T status) to FXII:C, FXII:Ag, and CRM classification, to resolve genotype–phenotype relationships.
2. **Prospective, adequately powered cohort** assessing thrombosis and pregnancy outcomes in genetically confirmed severe deficiency, with mandatory anti-FXII antibody and antiphospholipid testing to separate congenital from acquired low FXII.
3. **Functional reclassification pipeline** (HEK293T/CHO secretion assays + autoactivation kinetics, as in PMID: 41811038) to resolve VUS and distinguish loss- vs gain-of-function *F12* variants.
4. **Clinical-education intervention study** quantifying (and reducing) iatrogenic harm—unnecessary FFP, surgical delay, heparin-monitoring errors—triggered by isolated prolonged aPTT.
5. **Translational extension of garadacimab**: given the deficiency's safety, evaluate FXIIa inhibition in thrombosis, stroke, and extracorporeal circulation (dialysis, ECMO, cardiopulmonary bypass), where the congenital-deficiency phenotype predicts antithrombotic benefit without bleeding.

Report compiled from a 5-iteration autonomous investigation: 12 confirmed findings, 47 papers reviewed. Evidence spans human clinical, model-organism (mouse, cat), in vitro, and computational/evolutionary sources.